KILNDECK · PRODUCT REQUIREMENTS

KilnDeck product requirements and specifications

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DOCUMENT CONTROL

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Project TRL	TRL 3 Proof of concept		
Repository	github.com/BoujeeEnjinia1701/kilndeck		
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REVISION HISTORY

Version	Date	Author	Change
0.3	2026-10-28	Amish Chadha	Kit 1.8 migration to KND-PRS-001; R1 to R13 become PR-001 to PR-013 with tolerance, units and the user requirement each traces to; PR-014 (storm hold, KND-DEC-001 D12) and PR-015 to PR-017 (build, relaying and upkeep) added; R11 cost now USD 2,787 with the storm clamps
0.2	2026-10-03	Amish Chadha	R2 and R4 restated, R12 and R13 added (KND-DDR-001); status on paper from KND-CAL-001 for the constructable design
0.1	2026-09-30	Amish Chadha	Initial scaffold (as KND-REQ-001)

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On paper, the constructable design (option A, the rolling bridge deck) meets 13 of the 17 product requirements. PR-007 (floor temperature) is not met in full summer sun, PR-001 is met on the straight sides of the kiln but not on the curved ends, PR-008 is met only against an interim bearing limit until the partner kiln is surveyed, and PR-005 (dose repeatability) can only be judged by weighing doses at TRL 4. Every "met on paper" is a calculation in KND-CAL-001, not a test. Targets are proposals for review with a co-design partner. This document replaces KND-REQ-001; the old R numbers are in the "Was" column. Method: STANDARDS section 22.

Table 1. Product requirements and specifications.

ID	Traces to	Category	Title	Requirement	Target	Tolerance	Units	Verification	Status	Was
PR-001	UR-001, UR-003	Usability	No steps on the crust	The fireman shall feed every hole in a round without stepping onto the crust	0	max	steps per round	Observed trial over 20 rounds, TRL 4	Met on the straight sides; not met on the curved ends, which this prototype does not serve	R1
PR-002	UR-002	Safety	Guardrail	The guardrail shall have a top rail 1.1 m above the floor, a knee rail at 0.55 m and a 150 mm toe board, hold 1 kN at the top rail and take 1.5 kN without yield	1.1; 1.0; 1.5	min	m; kN; kN	Analysis (KND-CAL-001 s7), TRL 3; proof load (CalRig), TRL 4	Met on paper: 150 MPa at the proof load (yield 275 MPa)	R2
PR-003	UR-002	Safety	Rated load	The deck shall carry 250 kg and a proof load of 1.5 times that	250	min	kg	Analysis (KND-CAL-001 s6), TRL 3; proof load, TRL 4	Met on paper: 21 MPa in the chords at the proof load, with the carriage at mid-span	R3
PR-004	UR-004	Usability	Hand wheel force	Moving the loaded deck on level track shall take no more than 200 N at the hand wheel rim	200	max	N	Analysis (KND-CAL-001 s10), TRL 3; force gauge, TRL 4	Met on paper: 65 N level, 130 N on a 2 % grade	R4
PR-005	UR-006	Performance	Fuel dose	Each rotor pocket shall deliver the set dose of dry coal within 15 %	15	max	%	Weigh 20 doses, TRL 4	To verify at TRL 4; nominal 0.49 kg of coal per pocket	R5
PR-006	UR-003	Performance	Hopper capacity	The hopper shall hold at	1.0	min	rounds	Analysis (KND-	Met on paper: 52 L,	R6

ID	Traces to	Category	Title	Requirement	Target	Tolerance	Units	Verification	Status	Was
				least one feeding round for one fireman				CAL-001 s3), TRL 3	about 42 kg of coal against about 30 kg a round (1.4 rounds)	
PR-007	UR-005	Safety	Floor temperature	The standing area shall be at 50 °C or less after 6 h on an operating kiln	50	max	°C	Analysis (KND-CAL-001 s11), TRL 3; contact thermometer log, TRL 4	Not met in full sun: about 74 °C at 45 °C air; met at night (34 °C) and in shade (48 °C); the bare crust is about 150 °C	R7
PR-008	UR-007	Physical architecture	Load on the kiln	The pressure under each track pad shall stay below the limit set by a structural survey of the partner kiln	0.20	max	MPa (interim)	Analysis (KND-CAL-001 s9), TRL 3; survey and load cell, TRL 4	Met against the interim 0.20 MPa (0.151 MPa); cannot be confirmed until the survey	R8
PR-009	UR-003	Performance	Spout alignment	The spout outlet shall stop within 25 mm of the feed hole centre	25	max	mm	Analysis (KND-CAL-001 s12), TRL 3; measure at 10 holes, TRL 4	Met on paper: about 17 mm along the kiln, 10 mm across	R9
PR-010	UR-013	Safety	Parking brake	The lock pin shall hold the loaded deck on a 5 % grade with no creep	5	min	% grade	Analysis (KND-CAL-001 s10), TRL 3; inclined track test, TRL 4	Met on paper: holds 736 N against 491 N	R10
PR-011	UR-010	Cost	Cost	Value-engineering target for one deck and a demonstration length of track	3,500	target	USD	Costed BOM, TRL 3	Estimated cost of the constructable design: USD 2,787 (USD 713 under the target)	R11
PR-012	UR-008		Piece mass	No piece carried up the	40	max	kg	Analysis from the	Met on paper:	R12

ID	Traces to	Category	Title	Requirement	Target	Tolerance	Units	Verification	Status	Was
		Packaging and transport		kiln shall weigh more than 40 kg				model solids (KND-CAL-001 s5), TRL 3; weigh, TRL 4	heaviest is an end truss module at 39.8 kg; over the 33 kg team-handling guide (KND-DEC-001 D14, proposed, awaiting Amish)	
PR-013	UR-013	Safety	Track retention	The deck shall not run off a track end or drop off a track	0	max	derailments	Analysis (KND-CAL-001 s9, s17), TRL 3; push and lift tests, TRL 4	Met by design: end stops, buffers and derailment guards; the stops take 4.0 kN at a running speed of 0.28 m/s	R13
PR-014	UR-012, UR-013	Safety	Storm hold	Parked and clamped, the deck shall not move along the kiln in a 28 m/s gust	28	min	m/s	Analysis (KND-CAL-001 s15), TRL 3; pull test on the clamps and anchors, TRL 4	Met on paper with the storm clamps and tie-downs (KND-DEC-001 D12); anchor pull-out to assess at the partner kiln	
PR-015	UR-011	Materials	Local build	Every made part shall be cut, drilled and stick welded from stock steel sections, plate and sheet	0	max	special processes	Inspection of the model and build plan (KND-BLD-001), TRL 3	Met by design	
PR-016	UR-009	Packaging and transport	Track relaying	The track shall be laid ahead of the fire and lifted behind it	3,000	max	mm per module	Inspection (KND-BLD-001), TRL 3; relay trial, TRL 4	Met by design	

ID	Traces to	Category	Title	Requirement	Target	Tolerance	Units	Verification	Status	Was
				in 3 m bolted modules						
PR-017	UR-014	Usability	Upkeep	Every chain shall be guarded and every bearing a catalogue part that a local fitter can replace	0	max	special parts	Inspection of the BOM and model, TRL 3	Met by design	

Categories: physical architecture, performance, materials, usability, packaging and transport, safety. Status: met on paper, met by design, at risk, not met, over target, withdrawn, to verify at TRL 4.

Assumptions

- Kiln walls can carry the track pads without damage; to be confirmed kiln by kiln by a structural survey.
- The trench is about 6.0 m wide, the wall tops are at least 0.6 m wide and level with the crust within 100 mm, and feed holes are on a grid of about 1.0 m (KND-CAL-001, section 1).
- Firemen and owners accept a deck if it does not slow feeding.
- The track is laid in sections on the straight sides and moved with the fire.
- The deck is worked in winds up to about 7 m/s and clamped and tied down when a storm threatens (KND-CAL-001, section 15).